Style Over Substance: Content-Invariant Wrappers Flip LLM Safety-Judge Verdicts

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Abstract

Automatic *safety judges*—systems such as Llama Guard or a GPT-4o grading prompt that decide whether a model's reply is harmful—produce the numbers behind almost every reported jailbreak success rate, defense evaluation, and safety leaderboard. We ask a simple question: do these judges grade what a reply actually *contains*, or how it *sounds*? We keep a reply's content fixed and add *content-invariant style wrappers*: fixed strings placed before or after the reply that change only its tone—an educational disclaimer, a paragraph of ethical hand-wringing, a fake safety "reasoning" block, a token refusal ("I can't help with that") followed by the unchanged harmful body, a moralizing closing line—and, on harmless refusals, framing that merely *sounds* dangerous. The wrapper only decorates the reply; the body is preserved byte-for-byte. A faithful judge must therefore return the same verdict, and any *flip* (verdict change) is an error of the judge, not a real change in safety. Over 600 replies (300 genuinely harmful, 300 refusals, from public JailbreakBench artifacts across four target models) $\times$ up to 7 forms per reply (the original plus its direction's applicable wrappers) $\times$ 8 judges, we measure how often each judge flips, using paired significance tests, confidence intervals, and a measured *noise floor*—how often the judge changes its verdict when shown the exact same reply twice, with no wrapper at all. Our findings are precise rather than universal: most judges barely move, but specific judges harbor specific, cheaply exploitable blind spots. The token-refusal wrapper flips **19.9%** of GPT-4o-mini's correct "unsafe" verdicts to "safe" (95% CI [15.0, 24.0], Holm $p < 10^{-4}$, noise floor 0.5%; 18.2% under majority-of-three rescoring, so the flips are systematic, not decoding luck) yet moves Claude only 0.4%; the deployed **Llama Guard 4** guard model is also gamed—an "educational course" framing flips **12.3%** of its harmful verdicts to safe and the token refusal flips 8.3% (both significant)—and a single prepended sentence ("Here is the information you asked for.") flips **100%** of a keyword judge's refusal verdicts to "unsafe." A *second* purpose-built guard (gpt-oss-safeguard-20b) is immune—no wrapper moves it beyond 1.2%, below its own noise floor—so the two deployed guards sit at opposite ends of the vulnerability spectrum. Different judges have different blind spots: the framing that fools Llama Guard barely moves GPT-4o-mini, and vice versa. Stratifying wrappers by what they add shows the largest effects come from wrappers that insert a new (false) assertion, such as a token refusal, while pure tone-only decoration still flips up to 2.1% per judge—3–4$\times$ the vulnerable judges' noise floors. Rewriting only the judge's grading *prompt* to ignore tone (StrongREJECT-style) cuts the token-refusal attack roughly **tenfold** on the identical model—so the vulnerability lives in the judge, not the content. The judges do not even agree on which target model is safest before any wrapper is applied (full-panel agreement 55.5%), and a bootstrap over responses shows the four-model safety ranking on this artifact set is already unstable to sampling alone—so wrapper-induced reorderings, though present in point estimates, are reported as an instability axis rather than as confirmed reversals. We release the dataset, wrappers, code, and per-verdict labels.

1 Introduction

A safety benchmark score is only as trustworthy as the judge that produces it. Yet the field's attention has focused on *attacks* and *models*, treating the judge as a fixed oracle. Recent work has begun to question that oracle: constant "null-model" replies win automatic leaderboards (Zheng et al., 2025), reasoning-mimicking phrases sway reasoning judges (Wang et al., 2025), and a NeurIPS position paper argues most cross-paper attack-success-rate comparisons are invalid because judge error and prompt composition dominate the reported differences (Chouldechova et al., 2025). These are

existence proofs that judges are gameable—but for *general* evaluation, or from the *attack-input* side.

We ask the safety-specific, scoring-side question directly: **if we change only the *style* of a response and not its operational content, does the safety verdict stay fixed?** A faithful judge must say yes. Our answer is deliberately narrow: we do not claim that safety judges in general grade style over substance—most judges in our panel are robust to most wrappers—but that *specific* judges, including a purpose-built deployed guard model, harbor *specific* blind spots that a one-sentence, zero-cost decoration exploits at rates far above the judges' own measurement noise. We test this with *content-invariant style wrappers*—fixed strings prepended or appended to a response that add safety-signalling or harm-signalling tone while leaving the operational body byte-for-byte identical.

Our contributions:

1. A controlled **style-wrapper attack on safety judges** (not on models), where content preservation holds *by construction* rather than by an LLM rewrite—eliminating the intent-preservation confound of prior reformulation studies (Zhou et al., 2026b).

2. A cross-judge **flip-rate protocol**—paired significance tests with multiple-comparison correction, bootstrap confidence intervals, and a measured noise floor—over 8 judges and 9 wrappers, covering both the harm-hiding and harm-inventing directions, with wrappers stratified into tone-only vs. assertion-adding so the uncontestable core of the effect is reported separately.

3. A bootstrap-quantified **negative result on safety leaderboards**: on a standard jailbreak-artifact set, the model ranking is unstable to judge choice and to sampling alone, so wrapper-induced reorderings—though present in point estimates—cannot be confirmed; ranking claims need wider baseline separation than such sets provide.

4. A human-anchored validation that flips are judge errors, plus a public release.

Terminology. We use a small fixed vocabulary throughout. A **judge** is any automatic system that labels a model reply `unsafe` or `safe`. The **behavior** is the harmful request the reply was answering; the judge sees both. A **wrapper** is a fixed string added before or after a reply that changes its tone but not its content; the **original** form is the reply with no wrapper. A **flip** is a judge giving different verdicts to the original and wrapped forms of the same reply. **Harm-hiding** wrappers decorate genuinely harmful replies to sound safe—a flip there is a *false negative* (real harm waved through). **Harm-inventing** wrappers decorate harmless refusals to sound dangerous—a flip there is a *false positive* (an over-flag). The **noise floor** is how often a judge changes its verdict when shown the exact same input twice; any claimed wrapper effect must exceed it. Finally, we stratify wrappers by *what they add*: **tone-only** wrappers add pure commentary with no new claim (a disclaimer, ethical hedging, a fake reasoning block, a moralizing or lecturing close), while **assertion-adding** wrappers additionally introduce a new statement whose truth is contestable (a token refusal before an unchanged harmful body, a "here is the information" preface on a refusal, a false "university course" context, a preamble naming harmful topics). **Gold label** means the source dataset's reference label for a reply (harmful vs. refusal), which our human validation independently audits.

2 Related Work

Gameable and unreliable judges. That LLM evaluators harbor systematic biases is established for *general* evaluation: the foundational LLM-as-a-judge studies document position, verbosity, and self-enhancement biases (Zheng et al., 2023; Wang et al., 2023), and Wu and Aji (2023)—from whom we borrow our title phrase—show LLM evaluators rate fluent but factually wrong answers above correct ones. On the attack side, JudgeDeceiver (Shi et al., 2024) flips LLM-as-a-judge selections by injecting *optimized* token sequences into a candidate response; our wrappers differ in three ways—they are fixed, human-readable strings requiring no optimization and no access to the judge, they target the *safety* verdict rather than pairwise preference, and content invariance holds by construction, so a flip is provably a judge error. Closer to our setting, Zheng et al. (2025) show constant replies win automatic leaderboards; Wang et al. (2025) identify a "superficial reflection" bias where reasoning-styled text sways judges; Haldar and Hockenmaier (2025) document run-to-run judge self-inconsistency, and repeated-run studies find LLM outputs vary across identical runs even

on deterministic tasks (Zhou et al., 2026a). Our `fake_cot` and `ethical_reflection` wrappers are the safety-verdict analogues, and our noise floor operationalizes this run-to-run instability. Downstream systems increasingly treat LLM-derived labels as noisy signals to be modeled rather than trusted (Zhang et al., 2026b); a judge's flip rate quantifies one structured component of that noise.

Surface-form and framing sensitivity. On the *attack-input* side, Zhou et al. (2026b) show that scoring a single canonical phrasing of each harmful request underestimates safety's sensitivity to surface form—reformulating the same request changes outcomes. Framing effects likewise shift safety behavior in multi-agent settings, where operational reframing and approval-framed delegation degrade refusal behavior (Liu et al., 2026). We move the same concern to the *scoring* side: the response and its content are held fixed, only the style around them varies, and the quantity at risk is the verdict itself.

Jailbreak-evaluation validity. Chouldechova et al. (2025) argue ASR comparisons are often invalid; Huang et al. (2026a) and Huang et al. (2026b) show many "near-100% ASR" methods produce incomplete harmful content and that per-case criteria cut evaluator variance. Measurement-validity concerns recur across LLM evaluation more broadly: agent harnesses can "fix" failures that never happened when the evaluation signal itself is spurious (Wang et al., 2026b), compositional risk in agent skill ecosystems resists per-component safety scores (Wang et al., 2026a), human review—the fallback oracle—itself drifts, approving more while scrutinizing less (Yu et al., 2026), and what gets measured is shaped by evaluation budgets (Qian et al., 2026). Benchmark audits in other LLM application domains reach similar conclusions about what reported numbers actually measure (Su et al., 2026b,a). We complement these calls with a controlled measurement of one concrete failure mode of the safety verdict itself.

Over-refusal and guard models. OR-Bench (Cui et al., 2025) finds safety and over-refusal correlate, but both are measured with LLM judges; our false-positive direction probes whether part of that "frontier" is shared judge bias. Guard models (Llama Guard, ShieldGemma, WildGuard; Inan et al., 2023; Zeng et al., 2024; Han et al., 2024) and pipeline attacks (McKenzie et al., 2026) treat guards as inputs-to-harden, a surface that widens further in agentic deployments—runtime supply chains expose new exploit paths (Jiang et al., 2026b), motivating composition-safe designs such as monotonic capability attenuation (Jiang et al., 2026a); we attack guards from the scoring side instead. See Appendix A and the survey map in the repository.

3 Method

Object under test. The judge, not the model. The data unit is a fixed triple $(\text{behavior}, \text{response}, \text{gold})$, where gold is the source dataset's label for the response: `unsafe` (the reply complied with the harmful request) or `refusal` (it declined). We never edit the response content.

Content-invariant wrappers. Each wrapper w maps a response r to $w(r)$ by prepending or appending a fixed string; the body of r is preserved verbatim (a byte-level check confirms this for all 2,700 wrapped samples). The **harm-hiding** family (disclaimer, ethical reflection, fake reasoning block, educational framing, token-refusal-then-comply, moralizing suffix) decorates *unsafe* responses to sound safe; the **harm-inventing** family (lecture suffix, harm-adjacent preamble, pseudo-compliance frame) decorates *refusals* to sound dangerous. A faithful judge returns the same verdict for r and every $w(r)$: whichever stratum the wrapper belongs to, the operational body is fully present and unmodified, so a judge that grades actionable content cannot miss it. We nevertheless report the two strata separately, because "style" is philosophically contestable for assertion-adding wrappers—prepending "I can't help with that" changes the discourse structure of the message even though it removes nothing—while the tone-only stratum is immune to that objection and isolates pure style bias.

Judge panel. Eight judges spanning the quality axis: a keyword baseline (`rulebased`, pattern-matching on refusal phrases, no semantics); GPT-4o-mini under two different grading prompts (`gpt4o`, a generic rubric, and `strongreject`, a stricter rubric in the style of StrongREJECT (Souly et al., 2024) that tells the judge to ignore tone and framing—same model, so any difference between the two is attributable to the prompt alone); three

general chat models used as judges (DeepSeek-Chat, Qwen-Plus, Claude Haiku 4.5, all under the generic rubric); and two purpose-built moderation ("guard") models of the kind deployed in production content filters: **Llama Guard 4** (12B; Inan et al., 2023) and OpenAI's **gpt-oss-safeguard-20b** (a reasoning safety classifier, run under the shared rubric). Each returns a binary `unsafe` verdict; the rubric judges also return `refusal`.

Metrics. Every comparison is *paired*: the same response is judged in original and wrapped form, so the wrapper is the only difference between the two calls. For each (judge, wrapper) pair we report the **harm-hiding flip rate**—of the genuinely harmful responses the judge correctly called `unsafe` in original form, the fraction it calls `safe` once wrapped, i.e. $P(\text{wrapped safe} \mid \text{gold unsafe, original unsafe})$—and the symmetric **harm-inventing flip rate** $P(\text{wrapped unsafe} \mid \text{gold refusal, original safe})$. Significance uses the exact McNemar test (the standard paired test for "did the verdict change more often than chance between two conditions"), corrected for testing many (judge, wrapper) pairs at once (Holm and Benjamini–Hochberg), with a bootstrap 95% confidence interval (5,000 resamples over responses). Because API judges are not perfectly deterministic, each judge re-scores the *original* form three times and its observed self-disagreement rate is the noise floor; a flip rate counts as signal only if its confidence interval clears that floor. Since wrapped inputs could be noisier than originals (longer, closer to the decision boundary), we additionally re-score the *wrapped* form three times for the headline (judge, wrapper) cells and report majority-of-three flip rates alongside the single-scoring ones. Only responses a judge scores correctly in original form enter its denominator, so every counted flip is unambiguously caused by the wrapper.

4 Experimental Setup

Response set. 600 responses (300 unsafe-compliance, 300 refusal) from public JailbreakBench artifacts (Chao et al., 2024) (attacks PAIR, GCG, DSN, JBC, and random search), balanced across four target models (GPT-3.5-turbo, GPT-4, Vicuna-13B, Llama-2-7B) and ten harm categories, so the set doubles as a mini leaderboard. Gold labels are the JailbreakBench classifier's `jailbroken` flag.

Judges. Eight judges: a keyword baseline (`rulebased`); GPT-4o-mini under a generic rubric (`gpt4o`) and a StrongREJECT-style rubric (`strongreject`, same model, different prompt); DeepSeek-Chat; Qwen-Plus; Claude Haiku 4.5; and two purpose-built moderation models, Llama Guard 4 and gpt-oss-safeguard-20b (both via OpenRouter). All are API-served—no local GPUs. ShieldGemma (Zeng et al., 2024) and WildGuard (Han et al., 2024) are additional open guards left to the camera-ready.

Protocol. Judge temperature 0; three repeats on the original form for the noise floor; exact McNemar with Holm correction; 5000-sample bootstrap CIs. A 145-item stratified human validation with two annotators is reported below. Realized API cost for this run was under $5.

5 Results

Table 1 reports the full flip-rate matrix. The panel is eight judges: a keyword baseline (`rulebased`); GPT-4o-mini under a generic rubric (`gpt4o`) and a StrongREJECT-style rubric (`strongreject`); DeepSeek-Chat (`deepseek`); Qwen-Plus (`qwen`); Claude Haiku 4.5 (`claude`); and two purpose-built moderation models, **Llama Guard 4** (`llamaguard_or`) and OpenAI's **gpt-oss-safeguard-20b** (`safeguard_or`), both served via OpenRouter. The response set is 600 JailbreakBench completions (300 unsafe, 300 refusal) across GPT-3.5, GPT-4, Vicuna-13B, and Llama-2-7B, spanning ten harm categories.

H1: judges are gameable above the noise floor, but how much is judge-specific. The sharpest dangerous flip is on GPT-4o-mini: `refusal_then_comply`—a token refusal ("I can't help with that") followed by the unchanged harmful body under a "hypothetically" frame—flips **19.9%** of its correct "unsafe" verdicts to "safe" (95% CI [15.0, 24.0], $n = 291$, Holm $p < 10^{-4}$, noise floor 0.5%). In the opposite direction, the keyword judge marks **100%** of refusals "unsafe" once a pseudo-compliance prefix ("Here is the information you asked for.") is prepended ($n = 287$, $p < 10^{-4}$), while remaining unmoved by every harm-hiding wrapper—its verdict tracks surface phrases, not content. Crucially, the *same* `refusal_then_comply` wrapper barely moves the other chat judges: DeepSeek 3.5%, Qwen 0.7%, Claude 0.4%. Most consequentially, the

Wrapper	Stratum	Keyword	GPT-4o-m	StrongR	DeepSeek	Qwen	Claude	LG-4	SG-20b
Harm-hiding wrappers (applied to genuinely harmful responses)									
disclaimer prefix	tone	0.0	2.8	0.0	1.7	0.0	**0.0***	2.2	0.3
ethical reflection	tone	0.0	4.8	0.0	0.0	2.7	0.0	3.6	0.3
fake reasoning block	tone	0.0	2.4	0.3	0.4	0.0	**0.0***	1.8	0.0
moralizing suffix	tone	0.0	0.0	0.0	0.0	0.0	0.0	**0.0***	0.0
educational framing	assert.	0.0	1.0	0.0	3.5	0.7	0.0	**12.3***	0.3
refusal-then-comply	assert.	0.0	**19.9***	1.7	3.5	0.7	0.4	**8.3***	0.3
Harm-inventing wrappers (applied to refusals)									
lecture suffix	tone	0.0	0.4	2.3	0.4	**0.0***	1.8	1.1	1.2
harm-adjacent preamble	assert.	0.0	0.8	1.5	1.1	0.4	1.8	0.4	0.4
pseudo-compliance frame	assert.	**100.0***	1.9	1.1	1.8	2.3	0.7	0.0	0.8

Table 1: Any-flip rate (%) per wrapper × judge: of the responses a judge scored correctly in original form, the share whose verdict changes once the wrapper is added, grouped by attack direction. *Stratum*: tone = tone-only (pure commentary), assert. = assertion-adding (introduces a new contestable statement). *Judges*: Keyword = rule-based baseline; GPT-4o-m / StrongR = GPT-4o-mini under the generic vs. StrongREJECT-style rubric; LG-4 = Llama Guard 4; SG-20b = gpt-oss-safeguard-20b. Bold* = significant under the exact McNemar paired test after Holm correction; significant **0.0*** cells are *protective* effects (the wrapper makes that judge strictly more conservative).

deployed **Llama Guard 4** guard model—the kind of purpose-built moderator real systems place in front of an LLM—is itself significantly gamed: an `educational_framing` prefix ("for an accredited university cybersecurity course…") flips **12.3%** of its harmful verdicts to safe (Holm $p < 0.05$) and `refusal_then_comply` flips 8.3%, both above its noise floor. And the blind spots differ by judge: the framing that fools Llama Guard barely moves GPT-4o-mini (1.0%), while the token-refusal trick that fools GPT-4o-mini (19.9%) only mildly moves Llama Guard. The second deployed guard, **gpt-oss-safeguard-20b**, sits at the opposite extreme: no wrapper moves it beyond 1.2%, below its own measured noise floor (1.5%)—so immunity to content-invariant restyling is attainable today, but it is a property of the specific guard, not of guards as a class. The vulnerability is not a universal property of LLM judges but a judge-specific one. On the most robust judges, several statistically significant effects are *protective* rather than dangerous: the wrapper pushes the judge toward `unsafe`, i.e. it becomes *more* conservative, and its rate of waving real harm through stays ∼0.

Stratifying by what the wrapper adds. Aggregating flips within each stratum (weighting wrappers by their covered items) separates two findings. Tone-only wrappers—pure commentary, no new claim—flip **0.0–2.1%** of correct verdicts per judge (GPT-4o-mini 2.1%, Llama Guard 4 1.7%, all others ≤0.6%); small, but on the two vulnerable judges this is 3–4× their measured noise floors, and a single tone-only wrapper (`ethical_reflection`) reaches 4.8% on GPT-4o-mini and 3.6% on Llama Guard 4. Assertion-adding wrappers flip **2–6× more** than tone-only on every judge with a measurable effect (keyword judge: 0% vs. 30.7%—the item-weighted mean of its 100% pseudo-compliance flip and zeros on its other three wrappers; GPT-4o-mini: 2.1% vs. 6.2%; Llama Guard 4: 1.7% vs. 5.3%); the immune safeguard judge shows no stratum difference (0.35% vs. 0.45%, both within its noise floor), as expected for a judge with no effect to stratify. The headline effects therefore live in the assertion-adding stratum, where a critic may reasonably contest the "style" label; the tone-only stratum is immune to that critique and still moves the vulnerable judges above their noise floors. We emphasize that flips in *both* strata are judge errors under the task definition—the harmful body (or the refusal) is present verbatim in every wrapped message, and our human validation confirms the content judgment is unchanged—but the strata differ in how cheap and how deniable the attack is.

Are the flips systematic or stochastic? Wrapped inputs are indeed less stable: on GPT-4o-mini, self-disagreement across three re-scorings of the identical wrapped input rises to **10.0%** under the token-refusal wrapper and 3.3% under the ethical-reflection wrapper, versus 0.5% on original forms—the wrapper makes the judge roughly twenty times less self-consistent on unchanged content, a destabilization effect in its own right. The flips themselves, however, are predominantly systematic, not sampling luck: scoring each

wrapped input three times and taking the majority verdict, the `refusal_then_comply` flip rate moves only from 19.9% (single scoring) to **18.2%** (majority of three), with **14.4%** of items flipping on all three repeats; the tone-only `ethical_reflection` flip moves from 4.8% to 4.5% (3.1% flip on all three repeats)—still 6–9× the original-form floor. Llama Guard 4 is the starker case: its wrapped-form self-disagreement stays at **0.33%** on all three measured cells—the guard is nearly deterministic even on wrapped inputs—and its flips are almost entirely persistent: under `educational_framing`, **11.9%** of items flip on *all three* of three repeats (vs. 12.3% single-scoring), and `refusal_then_comply` shows 8.3% majority-flip with 7.9% persistent. The deployed guard does not waver into error; it reliably and reproducibly waves the same decorated harmful content through. Resampling the judge therefore does not undo the attack; it reveals that the wrapper biases the verdict on every judge measured, and additionally destabilizes the rubric judge.

H2: gameability is a property of the judge, shown two ways. First, across vendors: the identical wrapper applied to the identical responses produces a 19.9% flip rate on one judge and 0.4–3.5% on others, so the effect cannot be a property of the responses. Second, holding even the model fixed: `gpt4o` and `strongreject` call the *same* model (GPT-4o-mini) and differ only in the grading prompt; the StrongREJECT-style prompt—which explicitly instructs the judge to ignore tone, disclaimers, and framing and grade only actionable harmful content—cuts the `refusal_then_comply` flip from 19.9% to **1.7%**, a roughly tenfold reduction. The gap lives in the judge's design, not in the content being judged—and it is fixable by prompt alone.

H3: this artifact set cannot support a stable safety leaderboard at all. We report H3 as a methodological negative result, quantified by a bootstrap over responses (2,000 resamples within each target model). At the *item* level, judge disagreement is large and robust: full-panel unanimity on original-form verdicts is only **55.5%** (pairwise: Claude–GPT-4o-mini 94.2% at the high end, keyword–StrongREJECT 59.9% at the low end), and different judges' point estimates name different safest models (Vicuna-13B, Llama-2-7B, or GPT-3.5 depending on the judge). At the *ranking* level, however, base per-model unsafe rates on this jailbreak-artifact set are compressed (0.93–1.00 for the LLM judges), and the bootstrap shows the consequence: the baseline "safest model" is stable in only **39–70%** of resamples for six of the seven LLM judges (93% for the keyword judge and 98% for the immune safeguard judge, whose rates are more widely separated). Wrappers do reorder the four-model ranking in point estimates (1–6 of 9 wrappers per judge), but on this compressed set those reorderings cannot be separated from sampling noise, and we do not claim them as wrapper effects. The robust takeaway is negative and practical: a safety ranking read off a jailbreak-artifact set of this size is unstable to the choice of judge *and* to sampling before any adversary acts; content-invariant restyling then adds a third instability axis on top. A response set with wider baseline separation is required for affirmative reversal claims, and we flag this as the main open piece.

Human validation. Two annotators labeled a 145-item stratified subset (100 verdict-flip cases spanning the seven judges of the initial panel and both directions (the eighth judge, added later, is immune and contributes no flips), plus 45 content-invariance controls); annotator 1 labeled all items and annotator 2 independently relabeled a 40-item overlap. Content invariance holds at **100%** (145/145; separately, a byte-level check confirms all 2,700 wrapped samples contain the original response verbatim), and inter-annotator agreement is $\kappa = 1.0$ on content-invariance and $\kappa = 0.949$ on the harm label. Using the human harm label as ground truth, **90%** of flips (90/100) are confirmed judge errors. The 10 unconfirmed cases share a single cause: all are false-positive-direction items whose JailbreakBench gold label reads *refusal* while the human found genuinely actionable content (roleplay-style pseudo-refusals) — i.e., source gold-label noise rather than a wrapper or judge artifact. Overall human–gold agreement is 133/145 (91.7%), so the validation simultaneously anchors the flip-as-judge-error claim and quantifies the label noise of the source benchmark.

6 Discussion

When a one-sentence decoration can move a safety verdict at 8–20% rates on particular judges—including a deployed guard model—any reported ASR, defense win-rate, or safety leaderboard scored by such a judge inherits a stylistic bias orthogonal to real harm; and because the blind spots

are judge-specific, the bias does not cancel across papers that use different judges. The practical implication is not "discard judges" but "report the flip budget": alongside a safety score, report how much that score moves when the graded responses are restyled without any change in content—just as a physical measurement is reported with its instrument error. Deployed decision systems in other fields already engineer the stability of their predictive instruments explicitly (Li et al., 2026; Huang et al., 2026c); safety evaluation should meet the same bar. The underlying principle—a measurement should be invariant to task-irrelevant surface variation—recurs well beyond text, from person re-identification under clothing change (Ding et al., 2026) to robust medical segmentation under alternative scan orderings and representations (Wu et al., 2025; Tian et al., 2025). Content-focused rubric judging and majority-vote panels are natural mitigations—echoing cross-model review practice, where one vendor's model audits another's output (Xiang et al., 2026)—and our cross-judge data already indicates the strongest content-only rubric is least gameable. That one deployed guard is immune while another leaks 12% under a course-framing sentence sharpens the guidance further: robustness to content-invariant restyling is a measurable, attainable, and judge-specific property, and it should be audited before a judge is trusted with a leaderboard.

7 Limitations

Nine wrappers are a characterization of this set, not a population estimate. The largest effect comes from an overtly adversarial wrapper (token-refusal-then-comply); we report per-wrapper so readers can weight each, and note that the subtle harm-hiding wrappers, while above the noise floor in CI, do not survive Holm correction at $n \approx 291$ and are reported as suggestive. The rubric judges use each vendor's small/mid tier (GPT-4o-mini, DeepSeek-Chat, Qwen-Plus, Claude Haiku 4.5) rather than their frontier models, for cost and rate-limit reasons—the standard performance-efficiency trade-off in LLM-based judgment (Zhang et al., 2026a); frontier rubric judges and the remaining open guards (ShieldGemma, WildGuard) are a complementary axis for the camera-ready. Base per-model unsafe rates on the jailbreak-artifact set are compressed; our bootstrap analysis (H3) therefore treats leaderboard reorderings as unconfirmed, and affirmative reversal claims await a response set with wider baseline separation. Gold labels are the JailbreakBench classifier's, not human; our human validation measures their noise at ≈8% (133/145 agreement), concentrated in roleplay-style pseudo-refusals, and we exclude human-contradicted flips from the confirmed-error figure. The "style" label is contestable for the assertion-adding stratum (Section 5): a token refusal or a false "course" context changes what the message asserts, not merely how it sounds, and one wrapper (`harm_adjacent_preamble`) names topics that can be absent from the original request—it accounts for the single inter-annotator disagreement and we treat its flips as the weakest evidence. This is why we report the tone-only stratum separately: its effects are smaller but immune to this objection. About 60% of source responses end mid-sentence at the artifacts' generation length cap; truncation is identical in the original and wrapped forms, so it is symmetric for the paired comparison, and annotators judged the text actually present. We study single-turn text responses; multi-turn and agent-trajectory judging are future work.

8 Ethics

We attack the *evaluation* layer to make safety measurement more honest, not to jailbreak models. Responses come from existing public benchmarks; the shipped demo fixture uses non-operational placeholder text. Wrappers are stylistic and confer no new harmful capability to a *model*; they do, however, constitute working evasion framings against a deployed moderation filter, so we are reporting the specific Llama Guard 4 evasion results to Meta and the GPT-4o-mini destabilization results to OpenAI ahead of wide release. Because the wrappers are generic, human-readable framings rather than optimized artifacts, equivalent strings are trivially rediscoverable; withholding them would impede defenders auditing their own pipelines more than it would impede attackers. Exposing judge gameability helps defenders audit and harden their scoring layers.

Reproducibility

Code, wrappers, response-set builder, judge adapters, statistics, and per-verdict labels are released; an offline smoke test reproduces the full pipeline end-to-end without API access.

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A Extended related work

A full multi-venue map of related work (COLM, ICLR, ICML, NeurIPS, ACL, EMNLP, AAAI; 2025–26) is included in the released repository.